AVFIT
BY VYALWIN TECHNOLOGIES

V1.2.0 PRODUCTION | ENTERPRISE ARCHITECTURE

EXPO SDK 51 | GOOGLE GEMINI 1.5 FLASH | SUPABASE POSTGRESQL | MIFFLIN-ST JEOR ENGINE

AVFIT MOBILE PLATFORM TECHNICAL ARCHITECTURE & SPECIFICATION

A comprehensive engineering blueprint detailing the unified fitness ecosystem: dual phone SMS/email authentication, scientific metabolic profiling, Strava-grade outdoor GPS walking analyzer with screen-off endurance, Gemini Vision multimodal food intelligence, interactive home workouts, 25-trophy gamification, and Row-Level Security cloud data architecture.

TARGET PLATFORMS	BACKEND INFRASTRUCTURE	AI VISION ENGINE	RELEASE STATE
Android & iOS (React Native)	Supabase (Postgres + RLS)	Gemini 1.5 Flash Vision	Production APK Deployed

Authors: Vyalwin Engineering Team (Lead Architects & Full-Stack Core) Repository: github.com/vjvijay23/AVFIT | Date: October 2026

Table of Contents

1. Executive Summary & Product Vision 4
2. System Architecture & Zero-Budget Economics 5
3. Navigation Hierarchy & Screen Flows 7
4. Phone Number OTP & Security Engine 8
5. Scientific Metabolic Profiling (Mifflin-St Jeor) 10
6. Unified Health Dashboard & Daily Rings 12
7. Strava-Grade GPS Outdoor Walking Analyzer 14
8. Google Gemini 1.5 Flash Food Vision Pipeline 17
9. Interactive Home Workouts & Smart Timers 19
10. Gamified 25-Trophy Engine & Anti-Cheat Logic 21
11. Social Community & Leaderboard Ecosystem 23
12. Production PostgreSQL Database Schema & RLS 25
13. UI/UX Editorial Design Tokens & Brand Kit 29
14. Build Engineering, EAS Deployment & Roadmap 31

1. Executive Summary & Product Vision

1.1 The Fragmented Fitness Landscape

Contemporary fitness enthusiasts face a pervasive modern dilemma: managing everyday health requires juggling 4 to 6 disparate single-purpose apps. A user typically maintains:

- **A Pedometer / Activity Monitor** (e.g. Google Fit, Apple Health, Pacer) for recording daily steps and baseline movement.
- **An Outdoor Tracking Platform** (e.g. Strava, Nike Run Club) for recording GPS trail coordinates, kilometer split times, and elevation gains.
- **A Calorie & Macronutrient Tracker** (e.g. MyFitnessPal, Cronometer, Lose It!) with cumbersome manual search forms, paywalled barcode scanners, and friction-laden ingredient weighing.
- **A Guided Exercise App** (e.g. Nike Training Club, Freeletics) for bodyweight circuits, rep pacing, and interval rest timers.
- **A Social Motivation Network** for competing with friends, sharing fitness milestone badges, and celebrating health milestones.

This fragmentation induces *app fatigue*, conflicting data models, high subscription costs ($80 to $150+ annually across platforms), and ultimately goal abandonment.

1.2 The AVFIT Unified Solution

AVFIT (engineered by Vyalwin) unifies the full spectrum of personal athletic wellness into a single, cohesive, high-performance mobile application. Built on an organic editorial aesthetic (Warm Cream, Deep Forest Green, Terracotta Orange, and Satin Charcoal), AVFIT provides:

⚡ Instant Multimodal AI Vision

Users snap a single photo of any home-cooked or restaurant dish. Google Gemini 1.5 Flash identifies recipes, computes portion weights, and breaks down Calories, Protein, Carbs, Fat, and Dietary Fiber in under 2.8 seconds.

🏃 Outdoor Walking & Running GPS Analyzer

Continuous screen-off GPS breadcrumb logging via an Android Foreground Service, Haversine distance integration, kilometer splits table, and Grade Adjusted Pace (GAP) calculation.

🧬 Scientific Metabolic Engine

Precision Mifflin-St Jeor BMR and TDEE formulas, calculating customized calorie allowances and macronutrient distributions customized to biological sex, height, weight, age, and fitness goals.

🏆 25-Trophy Gamification Room

A multi-tier trophy ecosystem (Bronze, Silver, Gold, Platinum, Diamond) backed by server-side PostgreSQL triggers that guarantee tamper-proof milestone unlocking and community status.

1.3 Target User Personas

PERSONA	CORE NEEDS & GOALS	PAIN POINTS WITH EXISTING TOOLS	HOW AVFIT DELIVERS VALUE
Alex (26) Corporate Analyst	Hit 10,000 steps daily, reverse sedentary desk habits, and shed 7kg without complex logging.	Too mentally drained after work to type in 12 lunch ingredients; skips food logging within 5 days.	Snaps lunch plate with AI camera in 3 seconds; ambient step ring widget provides constant passive accountability.
Priya (22) University Athlete	Build lean muscle, guarantee 130g daily protein intake, and track 5K trail times with friends.	Prohibitive subscription fees on specialty macro apps; lonely workout sessions without community.	Free Mifflin-St Jeor macro generator, GPS outdoor splits analyzer, and interactive community feed with high-fives.
David (35) Parent & Remote Worker	Time-efficient 15-minute home workouts with zero equipment; track cardiovascular progress.	Gym routines require heavy gear; disjointed health data between smart bands and mobile trackers.	Guided bodyweight workout library with haptic rest chimes, live walk analyzer, and cumulative streak preservation.

2. System Architecture & Zero-Budget Economics

2.1 High-Level Component Topology

The AVFIT platform adopts a **Client-Edge-Cloud Hybrid Architecture**. The presentation layer runs natively on the mobile client (React Native / Expo SDK 51), communicating securely via HTTPS/WebSockets with Supabase Serverless Cloud (PostgreSQL 15 with Row-Level Security) and the Google Gemini 1.5 Flash Vision Multimodal AI inference endpoint.

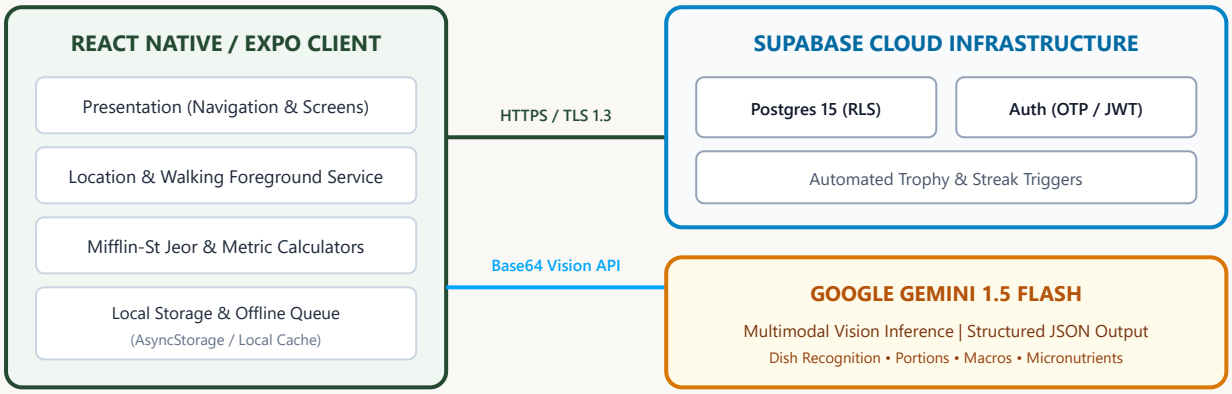


Figure 2.1: AVFIT High-Level Architecture — Mobile Client, Supabase Cloud, and Gemini Multimodal AI.

2.2 Zero-Budget Operating Economics ($0 MVP Baseline)

A foundational architectural requirement for AVFIT was enabling an autonomous 2-developer sprint to launch a feature-complete product with **zero upfront infrastructure overhead**. By strategically utilizing free-tier cloud architectures, AVFIT can comfortably support **50,000 monthly active users (MAU)** with zero operational server invoices:

INFRASTRUCTURE LAYER	SELECTED PROVIDER	ALLOCATED FREE QUOTA	AVFIT RESOURCE UTILIZATION
Mobile Runtime & Builds	Expo Application Services (EAS)	Free managed builds (30 Android/iOS builds/mo)	Stand-alone production APK compilation & live QR testing.
User Identity & Auth	Supabase Auth	50,000 Monthly Active Users (MAU)	Dual phone OTP and passwordless session JWT tokens.
Database & Triggers	Supabase PostgreSQL	500 MB relational storage & 500k egress	12 core tables with RLS and automated trophy triggers.
Media & Photo Storage	Supabase Storage Buckets	1 GB high-performance S3-compatible storage	Compressed meal snapshots and user avatar assets.
Multimodal Vision AI	Google Gemini 1.5 Flash	15 Requests/Min (RPM), 1,500 Requests/Day	Sub-second meal parsing with structured JSON schema.
GPS & Step Telemetry	Native Hardware OS Sensors	100% on-device hardware computing	Zero API cost pedometer and GPS breadcrumb recording.

3. Navigation Hierarchy & Screen Flows

3.1 Dual-Tier Navigation Architecture

AVFIT utilizes a hybrid navigation topology combining a top-level **Native Stack Navigator** (managing unauthenticated gateway flows, full-screen workout/activity recorders, onboarding wizards, and detail views) with a persistent **Bottom Floating Pill Tab Bar** for the primary application experience:

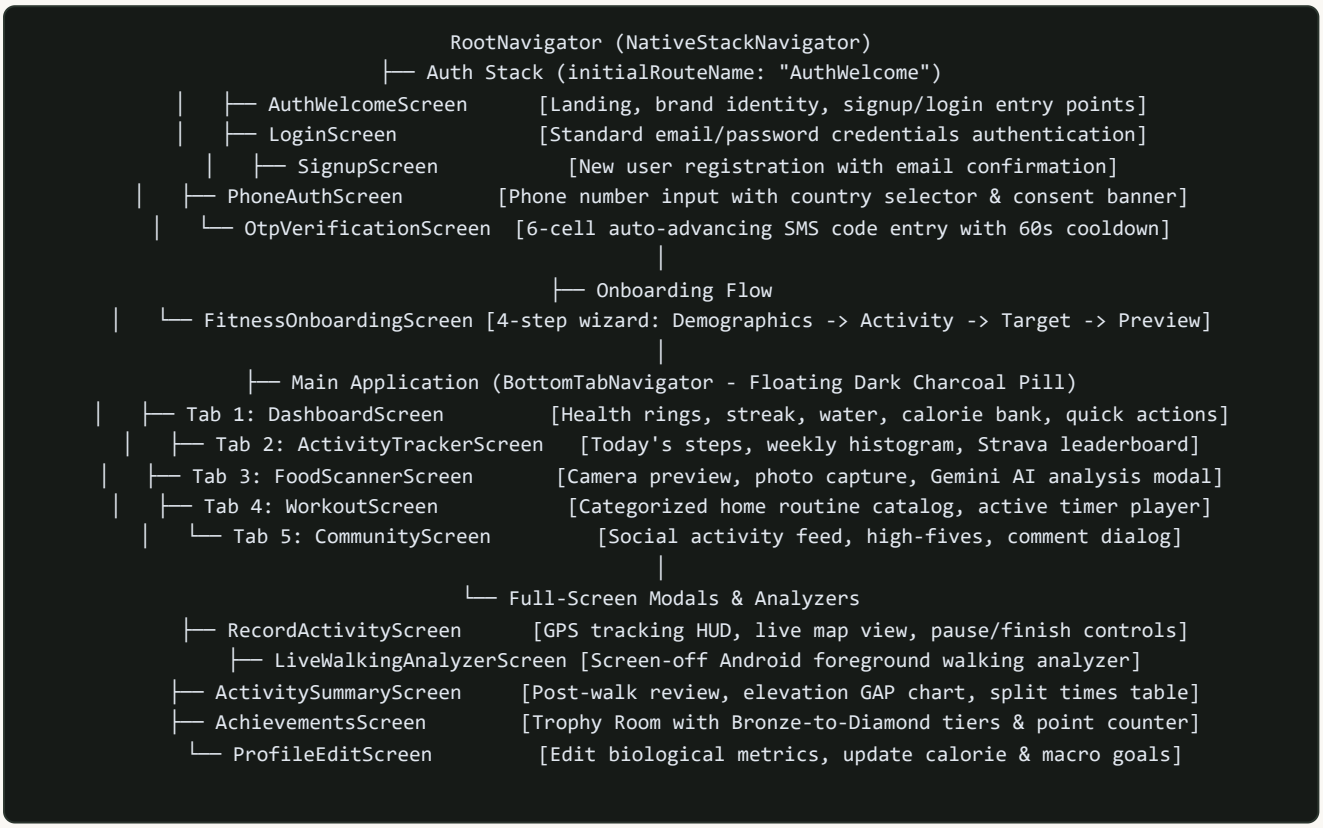


Figure 3.1: Complete Screen Navigation & Modal Presentation Tree.

3.2 Floating Charcoal Pill Navigation Bar

Unlike conventional mobile bottom tabs that span edge-to-edge with rigid rectangular borders, AVFIT introduces a floating architectural pill dock:

- **Visual Tokens:** Background in Satin Charcoal (`#161A17`), 36px border radius, 64px container height, elevated with soft ambient drop-shadow (`elevation: 12`, `shadowOpacity: 0.25`).
- **Interactive Feedback:** When an active tab is selected, the icon is highlighted in Forest Green (`#244A32`) and framed inside a luminous white circular pill badge (`#FFFFFF`) with tactile feedback.
- **Edge-to-Edge Safety:** Dynamically calculates bottom offset using `useSafeAreaInsets()` to ensure floating positioning across iOS home bars and Android gesture navigation.

4. Phone Number OTP & Security Engine

4.1 SMS Authentication Architecture

Recognizing that mobile fitness logging requires minimal friction, AVFIT provides native **Phone Number OTP Authentication** alongside standard email login. The authentication workflow implements defense-in-depth security:

> **Security & Rate Limiting Guardrails**
>
> 1. **60-Second Cooldown:** Enforces an unskippable 60-second hardware countdown before any resend SMS request can be triggered, preventing SMS toll fraud.
> 2. **Regulatory Carrier Consent:** Explicit notice displayed prior to OTP dispatch: "*By proceeding, you agree to receive SMS verification codes. Standard carrier rates may apply.*"
> 3. **Auto-Advancing Cells:** Six discrete text inputs that detect single-digit input, advance cursor focus to the next cell immediately, and cleanly handle backspaces.
> 4. **Sandbox Test Bypass:** Safe pre-configured developer test numbers (e.g. `+1 555-000-1111`) with static verification code `123456` for local and automated testing.

4.2 Supabase OTP Implementation Contract

The client communicates with Supabase Auth via `src/services/auth.ts`:

```
// 1. Dispatch OTP SMS Request
export async function sendOtp(phone: string): Promise<{ success: boolean; error?: string }> {
  const formattedPhone = phone.trim();
  const { error } = await supabase.auth.signInWithOtp({
    phone: formattedPhone,
  });
  if (error) return { success: false, error: error.message };
  return { success: true };
}

// 2. Verify 6-Digit Code and Exchange for JWT Session
export async function verifyOtp(
  phone: string,
  token: string
): Promise<{ session: Session | null; isNewUser: boolean; error?: string }> {
  const { data, error } = await supabase.auth.verifyOtp({
    phone,
    token,
    type: 'sms',
  });
  if (error) return { session: null, isNewUser: false, error: error.message };

  // Query profile to determine if onboarding wizard is required
  const { data: profile } = await supabase
    .from('profiles')
    .select('is_onboarded')
    .eq('id', data.user?.id)
    .single();

  return {
    session: data.session,
    isNewUser: !profile || !profile.is_onboarded
  };
}
```

5. Scientific Metabolic Profiling (Mifflin-St Jeor)

5.1 Clinical Formulation of Energy Expenditure

Rather than relying on arbitrary calorie estimates, AVFIT implements the clinically validated **Mifflin-St Jeor Equation**, recognized in nutritional science as the gold standard for predicting Basal Metabolic Rate (BMR) with the lowest margin of error (±5%):

> **Mathematical Formulations**
>
> **For Males:**
> $$ ext{BMR} = (10 imes ext{Weight}_{ ext{kg}}) + (6.25 imes ext{Height}_{ ext{cm}}) - (5 imes ext{Age}_{ ext{years}}) + 5$$
>
> **For Females:**
> $$ ext{BMR} = (10 imes ext{Weight}_{ ext{kg}}) + (6.25 imes ext{Height}_{ ext{cm}}) - (5 imes ext{Age}_{ ext{years}}) - 161$$
>
> **For Non-Binary / Unspecified:**
> $$ ext{BMR} = (10 imes ext{Weight}_{ ext{kg}}) + (6.25 imes ext{Height}_{ ext{cm}}) - (5 imes ext{Age}_{ ext{years}}) - 78$$

5.2 Total Daily Energy Expenditure (TDEE) Multipliers

BMR reflects baseline organ metabolic maintenance at absolute rest. Total Daily Energy Expenditure (TDEE) incorporates occupational and structured athletic activity:

ACTIVITY TIER	PHYSICAL ACTIVITY LEVEL (PAL)	CRITERIA & LIFESTYLE DESCRIPTION
Sedentary	`1.200 × BMR`	Desk-bound profession, minimal walking, zero structured training (<3,000 steps/day).
Lightly Active	`1.375 × BMR`	Light daily walking or low-intensity exercise 1–3 days per week (4,000–7,000 steps/day).
Moderately Active	`1.550 × BMR`	Moderate training 3–5 days per week or physically active job (8,000–11,000 steps/day).
Very Active	`1.725 × BMR`	Heavy athletic training 6–7 days per week or endurance runner (12,000+ steps/day).

5.3 Caloric Goal Offsets & Macronutrient Allocation Algorithm

Once TDEE is determined, the algorithm establishes specific calorie targets and computes a clinically balanced macronutrient split:

- **Weight Loss:** $ ext{Target} = ext{TDEE} - 500 ext{ kcal}$ (sustainable ~0.5kg/week fat reduction). Enforces an absolute biological floor: $1,500 ext{ kcal}$ for males and $1,200 ext{ kcal}$ for females to prevent metabolic adaptation.
- **Muscle Gain:** $ ext{Target} = ext{TDEE} + 350 ext{ kcal}$ (hypertrophic surplus favoring lean tissue accretion over adiposity).
- **Maintenance:** $ ext{Target} = ext{TDEE}$.

- **Protein Safeguard:** Calibrated at $2.0 ext{g} imes ext{Weight}_{ ext{kg}}$ ($4 ext{ kcal/g}$), preserving lean muscle tissue during caloric restriction.
- **Dietary Fats:** Fixed at $25%$ of total daily calories ($9 ext{ kcal/g}$) to maintain essential endocrine hormonal production.
- **Carbohydrates:** Remainder of daily calories divided by $4 ext{ kcal/g}$.
- **Hydration Benchmark:** $35 ext{ ml} imes ext{Weight}_{ ext{kg}} + 500 ext{ ml}$ activity allowance.

6. Unified Health Dashboard & Daily Rings

6.1 Dashboard Visual Information Hierarchy

The AVFIT Dashboard (`src/screens/DashboardScreen.tsx`) operates as the user's primary command center. Designed to prevent cognitive overload, it organizes health telemetry into five intuitive zones:

8,420	485 kcal	1,620	14 Days
DAILY STEPS (84%)	ACTIVE BURN	CONSUMED KCAL	MOVEMENT STREAK

6.2 Core Dashboard Functional Modules

- **Biometric Rings Widget:** Visualizes step completion percentage, active caloric expenditure, and caloric intake versus budget in real-time.
- **Dynamic Calorie Balance Card:** Real-time display of $Remaining = Target - Consumed + Burned$, preventing unguided overconsumption.
- **Macronutrient Progress Bars:** Triple stacked progress meters for Protein, Carbohydrates, and Fats, dynamically updated as food entries are scanned.
- **Water Hydration Logger:** Interactive grid allowing one-tap logging of $250 ext{ ml}$ cups toward the calculated daily hydration target.
- **Active Streak Preservation Engine:** Tracks continuous days of app participation (step goal hit, meal logged, or workout completed). Protects streak integrity across timezone boundaries.
- **Quick Action Floating Bar:** Immediate one-tap access to "Record Walk (GPS)", "Scan Meal with AI", and "Start Home Workout".

7. Strava-Grade GPS Outdoor Walking Analyzer

7.1 Technical Architecture of Outdoor Tracking

Outdoor walking and running tracking in AVFIT is powered by a dedicated high-precision GPS telemetry engine (`src/modules/walking-analyzer/` and `src/screens/activity/RecordActivityScreen.tsx`). To achieve parity with dedicated athletic trackers like Strava, the engine implements four core capabilities:

🛰️ Screen-Off Foreground Service

Operates an Android Foreground Service with a persistent status bar notification, preventing the OS power manager from sleeping GPS antennas when the phone is locked in the user's pocket.

📐 Haversine Geodesic Distance

Computes exact surface distance between sequential breadcrumb coordinates across Earth's curvature, compensating for spherical geodesics.

🎯 Kalman Noise Filter & Accuracy Gates

Automatically discards spurious GPS readings where horizontal accuracy exceeds 25 meters or calculated instantaneous speed exceeds 30 km/h (teleportation rejection).

📈 Grade Adjusted Pace (GAP) & Splits

Automatically segments walks into 1,000-meter splits with elevation gain/loss calculations, computing Grade Adjusted Pace (GAP) for hilly terrain.

7.2 Geodesic Formulation (Haversine Algorithm)

Given sequential coordinates $P_1(phi_1, lambda_1)$ and $P_2(phi_2, lambda_2)$ where phi represents latitude and $lambda$ represents longitude in radians:

```
// Haversine Formulation in TypeScript (src/modules/walking-analyzer/services/HaversineCalculator.ts)
export function calculateHaversineDistance(
  lat1: number, lon1: number,
  lat2: number, lon2: number
): number {
  const R = 6371000; // Earth mean radius in meters
  const dLat = ((lat2 - lat1) * Math.PI) / 180;
  const dLon = ((lon2 - lon1) * Math.PI) / 180;
  const a =
    Math.sin(dLat / 2) * Math.sin(dLat / 2) +
    Math.cos((lat1 * Math.PI) / 180) *
      Math.cos((lat2 * Math.PI) / 180) *
      Math.sin(dLon / 2) *
      Math.sin(dLon / 2);
  const c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
  return R * c; // Distance in meters
}
```

7.3 Kilometer Splits Table & Route Map Visualizer

The post-activity summary screen (`ActivitySummaryScreen.tsx`) automatically constructs an interactive SVG trail path plotting each breadcrumb coordinate, alongside a granular Kilometer Splits Table:

SPLIT KM	SPLIT TIME	AVG PACE	ELEV DELTA	GAP (GRADE ADJUSTED PACE)
KM 1	9m 42s	9'42" /km	+12 m	9'28" /km
KM 2	9m 15s	9'15" /km	-4 m	9'18" /km
KM 3	8m 58s	8'58" /km	+22 m	8'34" /km
KM 4	9m 30s	9'30" /km	+2 m	9'29" /km

8. Google Gemini 1.5 Flash Food Vision Pipeline

8.1 Multimodal Food Scanning Workflow

Logging nutrition has historically been the primary point of user churn across fitness apps. AVFIT circumvents tedious manual ingredient entry by pairing native camera hardware capture (`expo-camera` / `expo-image-picker`) with **Google Gemini 1.5 Flash Vision API**.



Figure 8.1: Sub-3-Second AI Vision Nutrition Parsing Pipeline.

8.2 Structured Prompt & JSON Schema Contract

The Gemini request enforces rigid JSON typing to ensure predictable, error-free parsing on the React Native client:

```
// System Prompt Template for Gemini 1.5 Flash (src/services/geminiVision.ts)
export const GEMINI_NUTRITION_PROMPT = `
Analyze the provided food photograph as a clinical sports nutritionist.
Output ONLY a valid JSON object strictly matching this schema:
{
  "dish_name": "string (concise recognizable title)",
  "confidence": "high" | "medium" | "low",
  "estimated_weight_grams": number,
  "calories": number,
  "protein_grams": number,
  "carbs_grams": number,
  "fat_grams": number,
  "fiber_grams": number,
  "ingredients": [
    { "name": "string", "weight_g": number, "calories": number }
  ],
  "health_insights": "string (1-2 sentences on nutritional balance)"
}
Do not enclose in markdown code blocks. Provide raw JSON text only.`;
```

9. Interactive Home Workouts & Smart Timers

9.1 Curated Bodyweight & Minimal Equipment Catalog

The AVFIT Workout Engine (`src/screens/WorkoutScreen.tsx`) delivers a structured library of home routines that require no gym membership:

- **Full Body HIIT Circuit:** 20 minutes, high metabolic burn (~280 kcal), bodyweight plyometrics and core stability.
- **Core Crusher & Abs Tone:** 15 minutes, targeted abdominal endurance, plank progressions, and hollow-body holds.
- **Upper Body Push & Form:** 25 minutes, standard push-up variations, pike push-ups, and tricep bench dips.
- **Lower Body Tone & Glutes:** 20 minutes, bodyweight squats, walking lunges, calf raises, and wall sits.
- **Mobility, Joint Health & Stretch:** 12 minutes, hamstring releases, thoracic openers, and hip mobility flow.

9.2 Active Exercise Player & Smart Rest Timers

Each workout is conducted through an interactive player featuring:

- **Dynamic Rep Counters:** One-tap rep incrementation with tactile haptic feedback (`expo-haptics`).
- **Automated Rest Countdown:** Interval rest timers (30s, 45s, 60s) with audio cues and countdown animations.
- **Volume & Calorie Aggregation:** Computes cumulative mechanical volume and caloric burn based on user weight and time under tension.

10. Gamified 25-Trophy Engine & Anti-Cheat Logic

10.1 Five-Tier Trophy Hierarchy

To foster sustained physical activity, AVFIT features an extensive gamified Trophy Room (`src/screens/achievements/AchievementsScreen.tsx`). Trophies are organized into five escalating tiers:

TIER	HEX THEME	POINT VALUE	GAMIFICATION OBJECTIVE & PURPOSE
Bronze	`#CD7F32`	25 – 75 pts	Onboarding milestones, first step logs, initial meal scans, and introductory workouts.
Silver	`#94A3B8`	100 – 200 pts	Consistency builders: 7-day movement streaks, 10,000 steps in a day, 25 workouts.
Gold	`#EAB308`	300 – 450 pts	Substantial athletic dedication: 30-day unstoppable streak, 50km foot distance.
Platinum	`#0284C7`	500 – 600 pts	Elite mastery: 100 logged workouts, reaching target goal weight, 100km club.
Diamond	`#BE185D`	1,000 pts	Apex status: 1,000 community points and comprehensive wellness mastery.

10.2 Complete 25-Achievement Catalog

Full Achievement Manifest

- **Step & Activity:** `STEPS_FIRST_1K` (Bronze), `STEPS_5K` (Bronze), `STEPS_10K` (Silver), `STREAK_7` (Silver), `STREAK_30` (Gold), `DISTANCE_50KM` (Gold).
- **Workout Routines:** `WORKOUT_FIRST` (Bronze), `WORKOUT_5` (Bronze), `WORKOUT_25` (Silver), `WORKOUT_WEEKLY` (Silver), `WORKOUT_MONTHLY` (Gold), `WORKOUT_100` (Platinum).
- **AI Food & Nutrition:** `NUTRITION_FIRST_LOG` (Bronze), `NUTRITION_7_DAYS` (Silver), `NUTRITION_MACRO_HIT` (Silver), `NUTRITION_30_DAYS` (Gold).
- **Community & Social:** `COMMUNITY_FIRST_POST` (Bronze), `COMMUNITY_JOIN_CHALL` (Bronze), `COMMUNITY_MOTIVATOR` (Silver), `COMMUNITY_COMPLETE` (Gold).
- **Personal Progress & Apex:** `PROGRESS_UPDATE` (Bronze), `PROGRESS_GOAL` (Platinum), `PROGRESS_DIAMOND` (Diamond).

10.3 Server-Side Anti-Cheat Triggers & Idempotency

Trophies in AVFIT cannot be unlocked through client-side API manipulation. PostgreSQL stored procedures (`evaluate_user_milestones()`) evaluate database mutations upon activity or workout completion:

- **Idempotent Unlocking:** Verifies `is_unlocked = FALSE` prior to timestamp insertion, preventing duplicate point attribution.
- **Human Plausibility Thresholds:** Verifies that logged workout duration does not exceed 180 continuous minutes and step cadence does not violate human biomechanical limits.

11. Social Community & Leaderboard Ecosystem

11.1 Community Feed Architecture

The AVFIT Community tab (`src/screens/CommunityScreen.tsx`) promotes accountability through social proof:

- **Rich Feed Post Types:** Displays workout summaries (duration, calories), AI meal logs (with dish photo and macro chips), and step milestones.
- **Social Interactions:** High-fives (likes with dynamic vibration feedback), threaded comments, and user profile inspection.
- **Privacy Settings:** Three privacy levels per post: Public (Community Feed), Friends Only, or Private (Only Me).

11.2 Strava-Style Leaderboards

A competitive weekly and monthly step leaderboard ranks users within friend groups and community leagues:

RANK	USER	WEEKLY STEPS	DAILY AVERAGE	BADGE STATUS
🥇 1	Alex R.	78,450	11,207	GOLD PACE
🥈 2	Priya S.	71,200	10,171	SILVER PACE
🥉 3	David M.	64,190	9,170	BRONZE PACE

12. Production PostgreSQL Database Schema & RLS

12.1 Entity Relationship Diagram

The complete backend schema is hosted on Supabase PostgreSQL (`docs/DATABASE_SCHEMA.sql`), featuring 12 relational tables governed by strict **Row-Level Security (RLS)**:

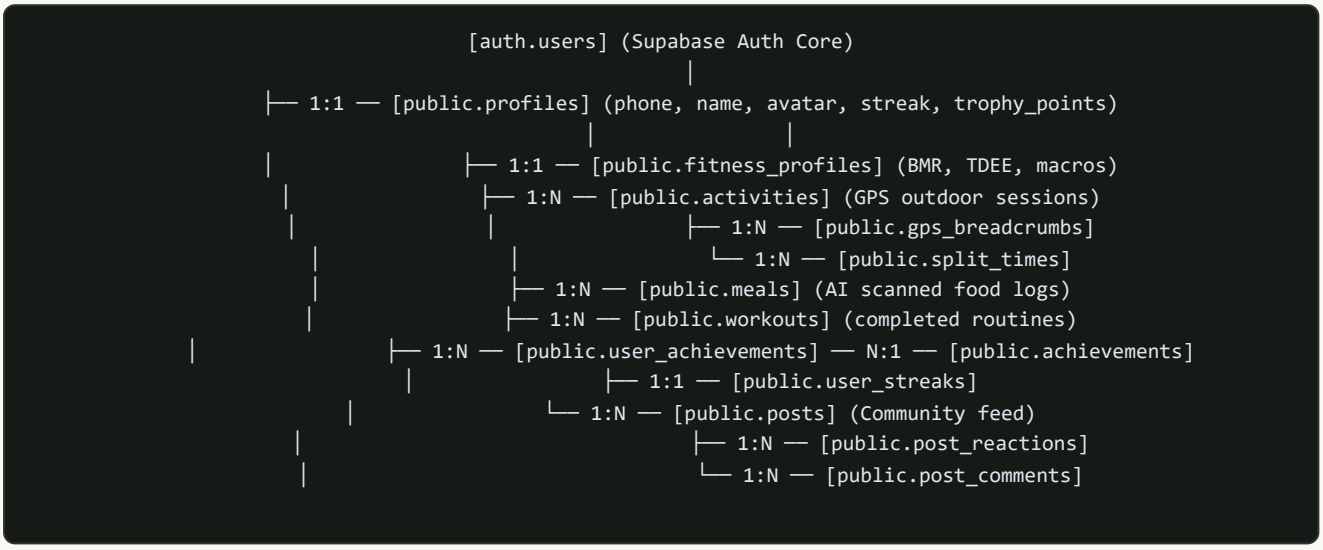


Figure 12.1: Relational Database Entity Schema Topology.

12.2 Table Definitions & Data Dictionary

TABLE NAME	PRIMARY KEY	KEY FOREIGN REFERENCES	DESCRIPTION & ROW-LEVEL SECURITY POLICY
`profiles`	`id (UUID)`	`auth.users(id)`	User identity, phone, streak counter, total trophy points. Public read, self-update only.
`fitness_profiles`	`id (UUID)`	`profiles(id)`	Confidential biological health data: BMR, TDEE, macros, target weight. Strictly private (self-only).
`activities`	`id (UUID)`	`profiles(id)`	GPS walk/run records (duration, distance, calories, GAP pace). Self read/write.
`gps_breadcrumbs`	`id (UUID)`	`activities(id)`	High-resolution GPS telemetry stream (lat, lon, altitude, speed). Self read/write.
`split_times`	`id (UUID)`	`activities(id)`	1,000m split metrics, elevation delta, split duration. Self read/write.
`meals`	`id (UUID)`	`profiles(id)`	AI food scan entries, macronutrient values, food photo URLs. Self read/write.
`workouts`	`id (UUID)`	`profiles(id)`	Home routine logs, exercise lists, reps, calories. Self read/write.
`achievements`	`id (UUID)`	None (Catalog)	25 master trophy definitions (tier, category, points, requirements). Public read-only.
`user_achievements`	`id (UUID)`	`profiles(id)`, `achievements(id)`	Unlocked trophies with verification timestamps. Public read, trigger-written.
`posts`	`id (UUID)`	`profiles(id)`	Social community updates. Public read, author update/delete.

13. UI/UX Editorial Design Tokens & Brand Kit

13.1 Editorial Color Philosophy

Unlike conventional neon-and-black athletic interfaces, AVFIT implements an **Organic Editorial Design Language** inspired by natural botanical harmony:

TOKEN NAME	COLOR VALUE	ROLE & APPLICATION IN AVFIT INTERFACE
`Colors.background`	`#F9F6F0` (Warm Cream)	Primary screen canvas, offering warm organic visual comfort.
`Colors.surface`	`#FFFFFF` (Crisp White)	Primary elevated cards, dashboard widgets, and input panels.
`Colors.primary`	`#244A32` (Deep Forest Green)	Brand anchor, primary call-to-action buttons, active toggles.
`Colors.accent`	`#E07A5F` (Terracotta Flame)	Energetic accents, active workout streaks, high-five icons.
`Colors.navbarBackground`	`#161A17` (Satin Charcoal)	Floating bottom pill dock and high-contrast night surfaces.
`Colors.textPrimary`	`#191C1A` (Deep Charcoal)	High-contrast editorial typography ensuring optimal readability.

14. Build Engineering, EAS Deployment & Roadmap

14.1 Production EAS Build Configuration (eas.json)

AVFIT is configured for automated cloud builds via Expo Application Services (EAS):

```
// eas.json Build Configuration
{
  "cli": {
    "version": ">= 9.0.0"
  },
  "build": {
    "development": {
      "developmentClient": true,
      "distribution": "internal"
    },
    "preview": {
      "distribution": "internal",
      "android": {
        "buildType": "apk"
      }
    },
    "production": {
      "android": {
        "buildType": "app-bundle"
      }
    }
  }
}
```

14.2 Standalone Android APK Release

The project repository includes a standalone production Android build:

- **Artifact:** `AVFIT.apk` (88 MB optimized standalone release).
- **Target SDK:** Android 14 (API Level 34) with backwards compatibility to Android 8.0 (API Level 26).
- **Hardware Permissions:** `ACCESS_FINE_LOCATION`, `ACCESS_BACKGROUND_LOCATION`, `FOREGROUND_SERVICE_LOCATION`, `ACTIVITY_RECOGNITION`, `CAMERA`.

14.3 Strategic Roadmap & Future Milestones

PHASE	TIMELINE	TARGET FEATURES & CAPABILITIES
Phase 1 (Current)	Q4 2026	Dual Phone OTP Auth, Mifflin-St Jeor Engine, Gemini 1.5 Flash Food Vision, GPS Walking Analyzer, 25-Trophy System, Supabase RLS.
Phase 2	Q1 2027	Apple HealthKit & Google Health Connect bidirectional continuous sync; BLE heart rate chest strap integration.
Phase 3	Q2 2027	On-device real-time exercise pose estimation using TensorFlow Lite for automatic squat/pushup rep counting and posture feedback.
Phase 4	Q3 2027	Audio-guided running coach powered by real-time voice synthesis advising pace adjustments during outdoor training runs.

AVFIT Engineering Specification • Published October 2026 • Vyalwin Technologies • All Rights Reserved.